

Ibraheem Aziz

Rawalpindi, Pakistan | ibraheemaziz3568@gmail.com | +92 332 8540999 | [LinkedIn](#) | [Portfolio: ibraheemaziz.me](#)

PROFESSIONAL SUMMARY

Computer Science student at FAST NUCES with hands-on experience in machine learning, deep learning, backend development, and software engineering. Experienced with Python, PyTorch, FastAPI, Docker, and RESTful APIs, with practical exposure to developing and deploying machine learning solutions.

TECHNICAL SKILLS

Languages: Python, C++, Java, Kotlin, JavaScript, HTML, CSS, x86 Assembly
AI & Machine Learning: Machine Learning, Deep Learning, PyTorch, Neural Networks, Regression, Classification, Clustering, Reinforcement Learning, Embeddings, Similarity Search
Core CS Concepts: Data Structures, OOP, Operating Systems, Computer Networks, Secure Coding
Backend & Databases: FastAPI, RESTful APIs, PHP, SQL, MySQL
Tools & Frameworks: Docker, Git, GitHub, Android Studio, SFML, OpenGL/GLUT

EXPERIENCE

DiveDeepAI

July 2026 – September 2026

ML Engineering Intern

- Worked on machine learning and deep learning workflows, including model development, evaluation, and integration.
- Developed ML-focused backend services using Python and FastAPI.
- Worked with PyTorch, pretrained models, embeddings, and similarity-based machine learning systems.
- Used Docker for containerizing machine learning applications and supporting deployment workflows.
- Applied machine learning concepts including regression, classification, clustering, neural networks, and model evaluation.

NCRA-UAV Dependability Lab

July 2025 – Aug 2025

Backend & Cybersecurity Intern

- Contributed to backend development and testing of a UAV monitoring and simulation system in a research environment.
- Implemented and evaluated path planning approaches including SARSA, A*, and Q-learning for autonomous navigation scenarios.
- Worked on battery management, multi-agent coordination, and collision avoidance mechanisms.
- Assisted in the implementation of cybersecurity attack detection and mitigation functionalities for system resilience.
- Contributed to system logging, monitoring, and performance optimization activities.

PROJECTS

Socially – Instagram Clone

Android, Kotlin, PHP, MySQL

- Developed a full-stack social media application replicating core Instagram features including feeds and posts.
- Designed UI/UX using XML and implemented app logic utilizing Kotlin within Android Studio.
- Built backend services using PHP and integrated MySQL for efficient user data management.

Aerologix MRO – Aircraft Maintenance System

Java Swing, PHP, MySQL

- Developed a desktop app to manage aircraft maintenance, repair, and overhaul operations.
- Implemented backend logic for service requests and inventory tracking, with a Java Swing front-end interface.

Troy – AI Voice Assistant

Python, OpenAI API

- Developed a desktop voice assistant using speech_recognition and pyttsx3 for automated audio processing.
- Integrated OpenAI API to process complex user queries and execute dynamic automated web navigation tasks.

Xonic Game & Arcade Clones

C++, SFML, OpenGL/GLUT

- Engineered a multiplayer platform utilizing a Min Heap for leaderboards and priority queues for fair matchmaking.
- Designed 2D games including Snake, Wordle, and Candy Crush clones using OOP concepts, SFML, and OpenGL.

EDUCATION

Bachelor of Science in Computer Science

Expected: Aug 2027

National University of Computer and Emerging Sciences (FAST NUCES) - Islamabad, Pakistan

Discipline: BS(CS)

Pre-Engineering (Intermediate)

Graduated: June 2023

Army Public School and College System - Rawalpindi, Pakistan

LEADERSHIP & COMMUNITY ENGAGEMENT

Community Computing Initiative*Volunteer Computing Instructor*

- Conducted a computing lab introducing basic programming using Scratch to local students.
- Fundraised and provided essential resources while delivering hands-on sessions to build foundational computer skills.